Advanced Embedded Architectures and Applications

Introduction

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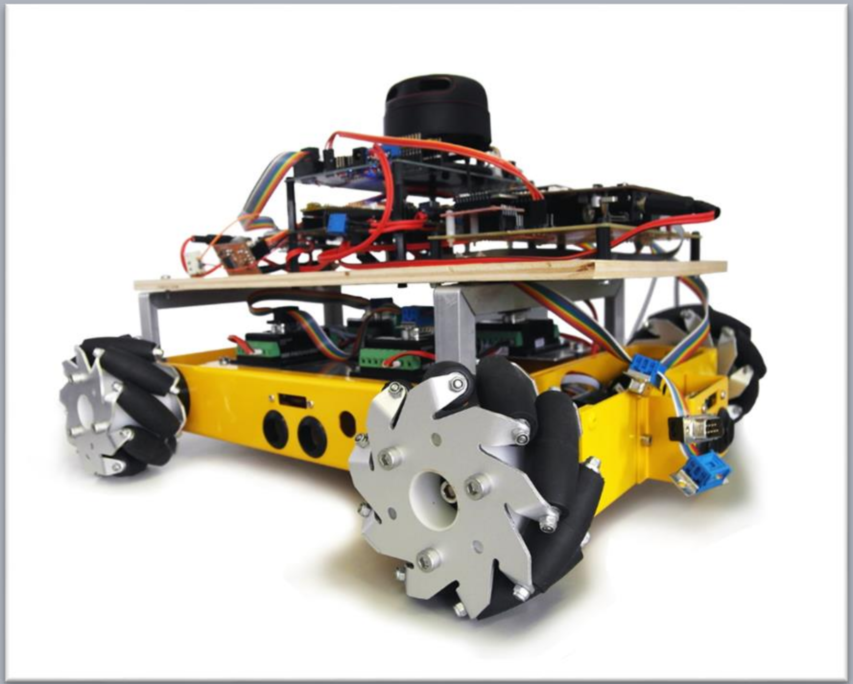


Content

- Course Goal
- Content
- Lab Organisation

Course Content

Lecture

- introduction to multitasking concepts and operating systems,
- structure and functionality of selected industrial embedded Operating Systems
- design of reactive systems, state machine design and coding,
- architectural development of embedded, realtime, multitasking systems
- analysis of embedded industrial architectures and design patterns (Basic Software, Application Software, Runtime Environment)
- automotive architectures, AUTOSAR
- embedded control system design
- multicore architectures
- safety architectures

Lab

- Separating basic software and application software introducing a runtime environment
- Developing a simple multithreading, reactive application

Learning outcome

to understand:

- the functionality of embedded operating systems
- the challenges and risks of multithreading architectures
- the structure of multicore controllers
- key design patterns of industrial embedded architectures

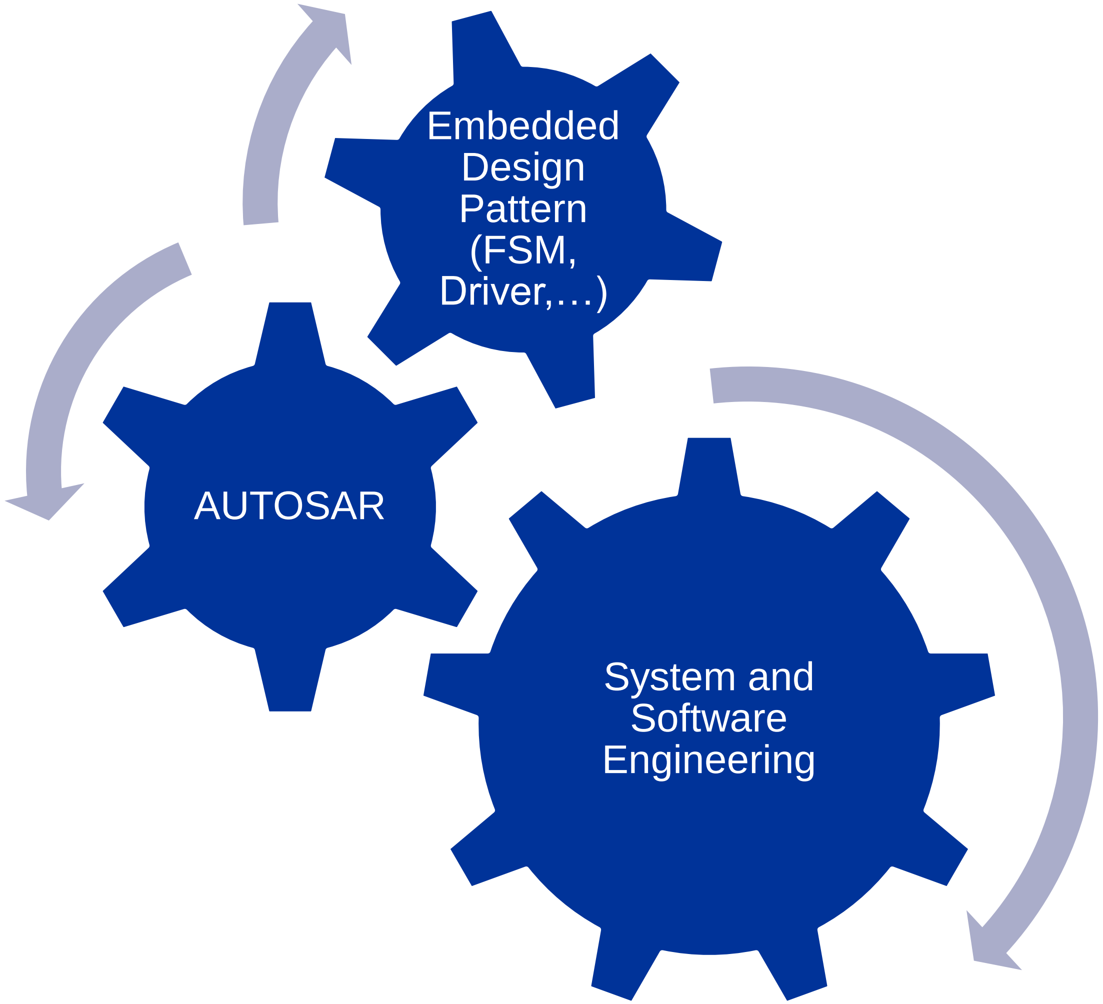
to apply:

- the gained knowledge to implements tasks and intertask communication on embedded controllers
- design and implement flat statemachines
- review, test and debug multithreading applications
- separate base and application software using the concepts of embedded runtime environments

to transfer:

- the design patterns and concepts to more complex embedded architectures using new operating systems and controllers

Big Picture



Embedded Design Pattern (FSM, Driver,…)
AUTOSAR
System and Software Engineering

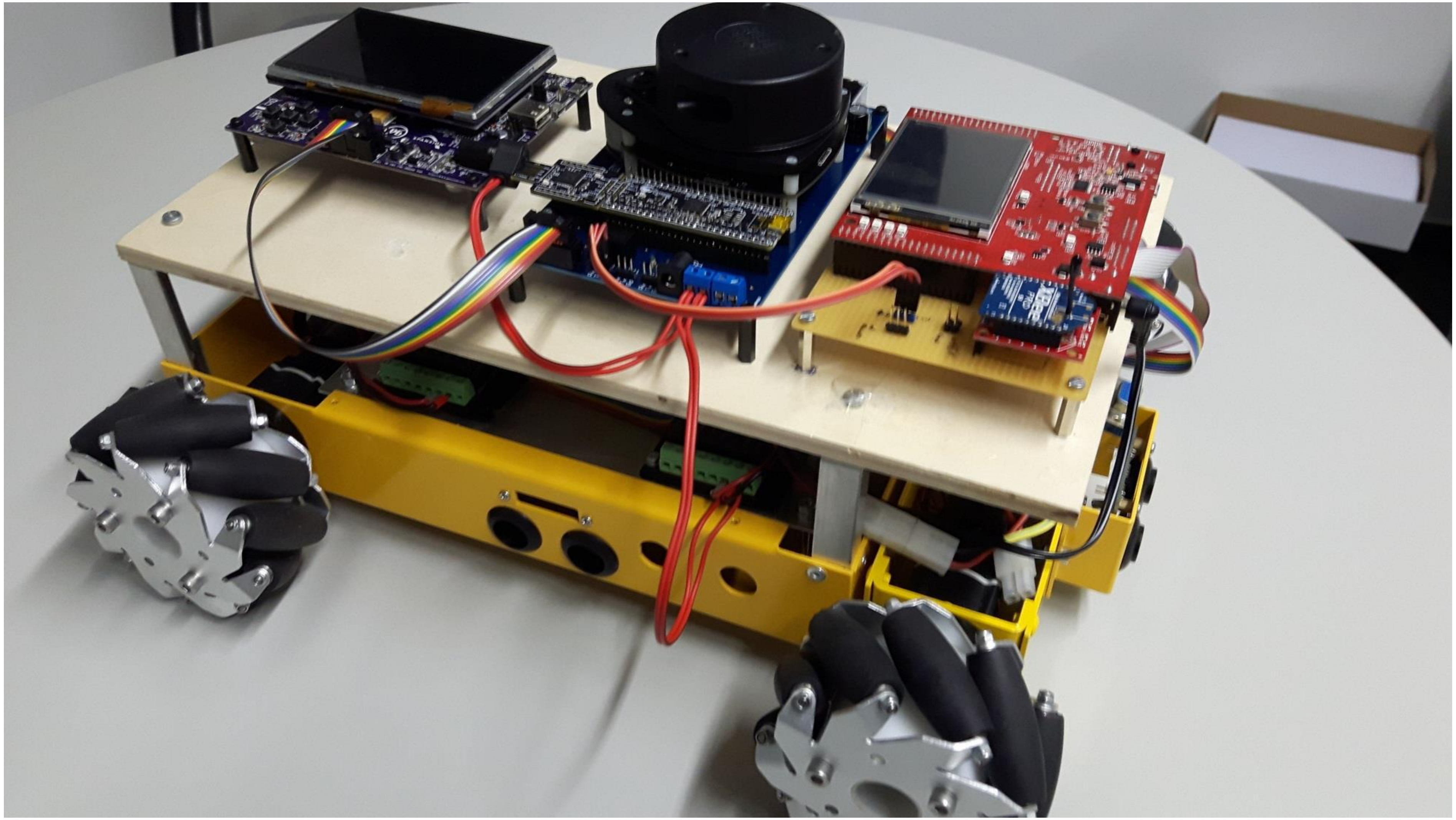
Structure of the Lecture – still draft!!!

Lecture	Content
1	System Engineering – Developing the Showcase
2	Realtime and Timing
3	Layered Architectures – the AUTOSAR concept
4	MCAL driver development (synchron, asynchron, buffering…)
5	Engine Control – Basics and Complex Device Driver Design
6	State Machine Design I – Lookup Table
7	State Machine Design II – Active Objects
8	Communication Protocols - ZigBee
9	Application Development

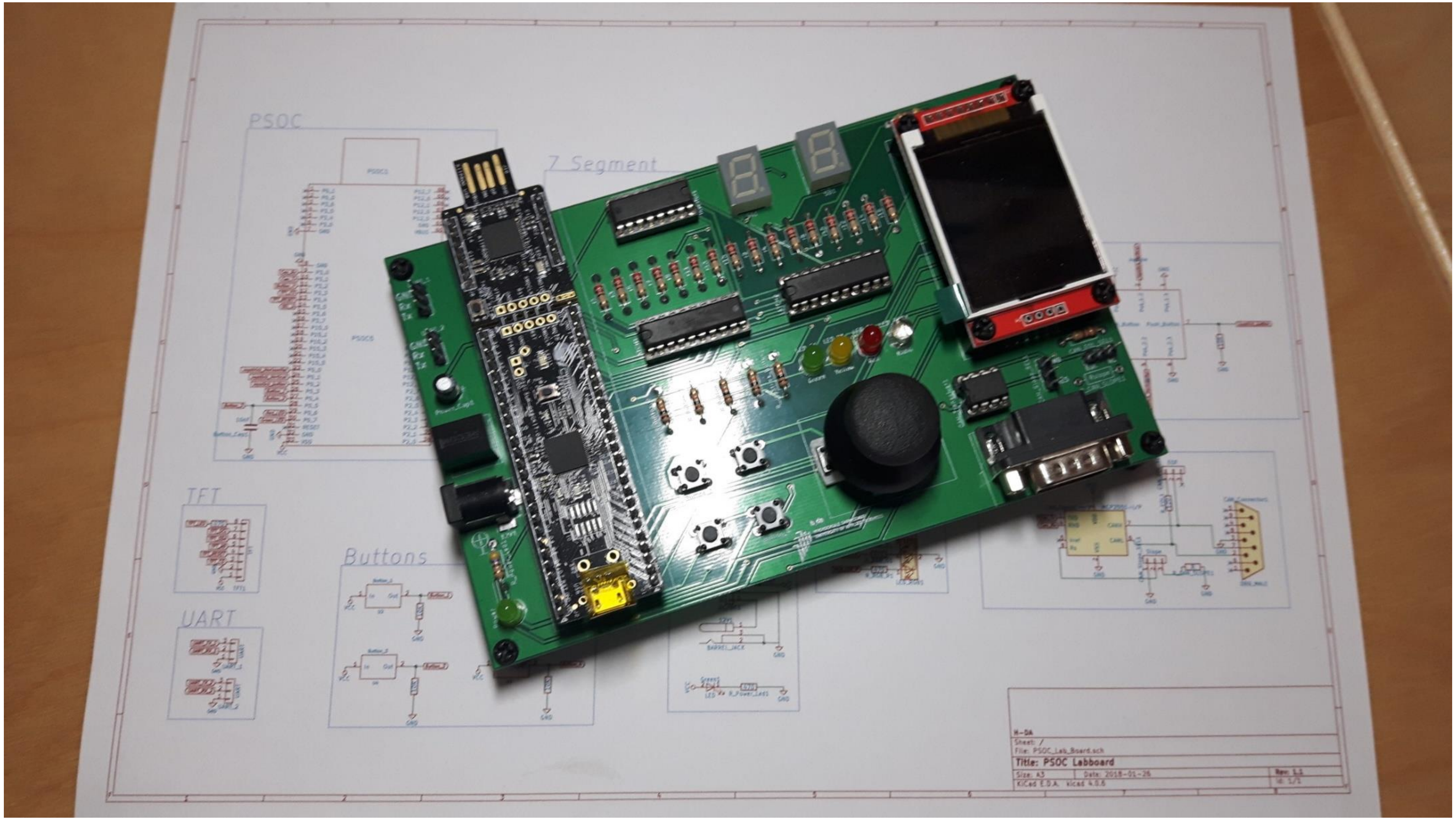
Structure of the Lecture – still draft!!!

Lecture	Content
10	Refactoring
11	Sensors / Filters
12	Debugging Concepts – Debugger, Data Tracing, Execution Tracing
13	Multicore

Toys for big boys and girls



LabBoard



Lab Organisation

Mandatory assignments

- AUTOSAR RTE Gaspedal
- Complex FSM Clock

Anybody without Labboard?

Lab execution

- Design and implementation as lab@home
- Labs will be reviewed and must be passed (exam prerequisite)
- Lab dates (reviews) will be communicated

Lab content will be part of the exam!